Split-Zero Geometry and Common Deformation Registers

Project atlas, exact results, formal checks, and visual work

The Clankers

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Contents

1 Read this first

This record is the public home for a continuing mathematics project at the intersection of explicit arithmetic registers, the Erdos–Strauss research programme, split-zero and tagged-sedenion models, finite moonshine and $\widehat{Z}$

calculations, and exact geometric experiments. It is organised as a working mathematical library, not as a sequence of announcements.

The concept DOI is **10.5281/zenodo.20822444**. It resolves to the latest version. New releases belong to this concept and do not create a second project page.

The public surface has three layers:

1. This atlas states what is established, what is generated, what has been obstructed, and what is genuinely open.
2. The results compendium contains the complete current packets, including formulae, finite tables, proof statements, and source boundaries.
3. The ZIP archives preserve Lean, Python, ledgers, data, editable visual sources, and machine-readable text. They are evidence and replay material, not the front door.

No JT-authored corpus dump is included. Jacolm Tobley (JT) remains credited as a researcher and project relationship. Direct source material remains under its own provenance and ownership. The public payload consists of work supplied by the repository maintainer, generated project mathematics, exact checks, and rights-screened derivatives. The release licence is CC BY 4.0; third-party literature keeps its original copyright and is cited rather than republished.

1.1 File map

File	Use it for
`00_PROJECT_ATLAS_20260717.pdf`	Human front door, mathematical map, status boundaries, and reading routes.
`01_PROJECT_ATLAS_20260717.md`	Searchable and reusable source of the atlas.
`02_CURRENT_RESULTS_COMPENDIUM_20260717.pdf`	Complete current result packets with bookmarks and a contents index.
`03_FORMALIZATION_AND_EXACT_CHECKS_20260717.zip`	Lean sources, Python checkers, ledgers, build/audit logs, and replay guide.
`04_VISUAL_ATLAS_20260717.pdf`	Selected exact and explanatory visual results.
`05_VISUALIZATIONS_AND_DATA_20260717.zip`	Editable visual sources, scene data, GeoGebra/HTML assets, renders, and QA.
`06_CURRENT_WORKING_TEXTS_20260717.zip`	TeX, Markdown, plain text, and per-packet machine extracts.
`README.md`	Short navigation and citation instructions.
`CHANGELOG.md`	Version history and material changes.
`PROVENANCE_AND_RIGHTS.md`	Inclusion, exclusion, credit, and reuse rules.
`MANIFEST.json` / `SHA256SUMS.txt`	Machine-readable inventory and integrity checks.

1.2 Status vocabulary

The record uses the following labels literally.

Label	Meaning
Exact theorem	A finite or symbolic statement proved in Lean, or an exact calculation replayed independently with a complete finite certificate.
Checked calculation	A reproducible calculation whose stated finite or bounded scope has been replayed.
Generated construction	A mathematically exact object built by the project, but not recovered as the unique or source-defined object.
Obstruction / no-go	A tested candidate map or identification fails for an explicit reason or counterexample.
Diagnostic	A deliberately artificial or degenerate model used to expose an underconstrained interface.

Label	Meaning
Evidence-backed open problem	A positive pattern survives nontrivial checks, but a named proof, source, or naturality step remains missing.
Investigation prompt	A similarity worth testing. It is not a conjecture and carries no assertion of truth.
Historical exploration	Preserved provenance that is superseded, false, under-specified, or no longer part of the current result surface.

A statement is not promoted to a conjecture merely because it appeared in a conversation or because a user gestured toward it. Conditional statements invented to avoid saying that a proposed map is false are excluded from the current result surface. Failed maps are recorded as failed maps.

2 Arithmetic register and the Gamma-107 chain

The clearest new theorem-level chain begins with a finite arithmetic register and ends with an exact generated order-six target model. The complete Lean statements are exposed through `FiniteRegisterGamma107PublicFacade.lean`; the implementation tables and replay evidence are in the formalization archive.

2.1 The 48-state register

Let U_{24} denote the unit classes modulo 24. The checked character map

$$\Phi : U_{24} \longrightarrow \mathbf{F}_2^3$$

is bijective and multiplicative. The six accepted conductor faces are represented by

$$S_{840} = \{1, 121, 169, 289, 361, 529\}.$$

The accepted square-support register $G_{840}^{\square 35}$ has exactly 48 elements and decomposes as an explicit product of an eight-element ground-character register with a six-sector coordinate. Its square-class map has the checked kernel described in Packet 129.

Public theorem: `acceptedFiniteRegisterIsComplete`.

Status: exact finite theorem. The public alias uses standard Lean `propext`; it uses no native evaluator axiom.

2.2 Tagged sedenion preflight

The target preflight uses a 32-basis tagged algebra with multiplication modelled as $A_4 \otimes \mathbf{R}[C_2]$. The packet checks the basis product, the cross-polarised zero-divisor pairs, the quartet squares and associators, and the block data of the displayed operator Γ_{107}.

The exact block dimensions are

$$\dim V_0 = 8, \qquad \dim V_2 = 16, \qquad \dim V_4 = 8,$$

with characteristic and minimal polynomials

$$\chi_{\Gamma_{107}}(x) = x^8(x^2+4)^8(x^2+16)^4, \qquad m_{\Gamma_{107}}(x) = x(x^2+4)(x^2+16).$$

Public theorem: `gamma107BlockDecomposition`.

Status: exact finite theorem. This is a checked algebraic model, not an authorship claim about the source corpus.

2.3 Sector 289 and the unnormalised no-go theorem

Multiplication by 289 on the 48-state register produces eight disjoint six-cycles. On the target, the sixth power of the unnormalised operator has diagonal scale factors

$$0 \text{ on } V_0, \qquad -64 \text{ on } V_2, \qquad -4096 \text{ on } V_4.$$

No nonzero vector is fixed. Therefore an exact intertwiner from the tested period-six source action to the unnormalised Γ_{107} action must vanish.

Public theorems: `sector289ActsByEightSixCycles` and `unnormalisedGamma107Sector289NoGo`.

Status: exact obstruction. It rules out this unnormalised target action; it does not rule out a normalised, projective, rescaled, or exponential action.

2.4 The kernel-collapse diagnostic

An artificial action can be made equivariant by forcing its image into the kernel of Γ_{107}. The resulting code can still be injective at the finite interface level. This proves that the old bridge specification was too weak: interface inhabitance did not exclude a mathematically degenerate model.

Public theorem: `degenerateKernelBridgeDiagnostic`.

Status: diagnostic counterexample. It is not a candidate bridge, not a source result, and not evidence for the desired naturality.

2.5 Generated exponential order-six repair

Write $G = \Gamma_{107}$ and define the spectral projectors

$$P_0 = \frac{(G^2+4I)(G^2+16I)}{64}, \qquad P_2 = -\frac{G^2(G^2+16I)}{48}, \qquad P_4 = \frac{G^2(G^2+4I)}{192}.$$

The generated target step is

$$E = P_0 + \left(\frac{1}{2}P_2 + \frac{\sqrt{3}}{4}GP_2\right) - \frac{1}{2}P_4 + \frac{\sqrt{3}}{8}GP_4.$$

Exact arithmetic gives

$$E^6 = I, \qquad E^k \neq I \ (1 \le k < 6), \qquad \chi_E(t) = (t-1)^8(t^2-t+1)^8(t^2+t+1)^4.$$

The project constructs 48 distinct nonzero target codes with exact equivariance, a checked alternating parity reader, and a fixed provenance reader. On a frequency-two plane the six phases are the exact $\mathbf{Q}(\sqrt{3})$ hexagon

$$(2,0), (1,1), (-1,1), (-2,0), (-1,-1), (1,-1),$$

under the pair update

$$(a,b) \longmapsto \left(\frac{a-3b}{2}, \frac{a+b}{2}\right).$$

Public theorem: `generatedExponentialC6Repair`.

Status: exact generated construction. The finite phase model is proved; the analytic exponential motivation is checked separately. No source-defined normalisation or source-natural bridge has been recovered.

Genuine open problem: `SourceNaturalNondegenerateBridgeExists` asks for a nonzero bridge for an externally supplied source action. The proposition is defined but not inhabited.

3 Moonshine, E8 cubed, and Sigma(2,3,5)

The second main body of work compares exact q-series and finite module data between an E_8^3 umbral-moonshine surface and sectors associated with $-\Sigma(2,3,5)$. The compendium preserves the source crosswalk in Packet 76 and the complete sequence of Packets 77–108 and 115–120 because the negative results are as important as the surviving equalities.

3.1 Exact common output series

The project identifies exact shared coefficient series χ_0 and χ_1 at the displayed output level. Named endpoint pullbacks and finite labels make the comparison explicit and reproducible.

Status: checked common output data. Equality of displayed series does not imply an isomorphism of modules, manifolds, defects, or vertex-algebra objects.

3.2 Internal-morphism boundary

The source-backed cone-VOA construction supplies a rigorous method dependency, but the displayed ranks, symmetry actions, topology, and internal carriers are not the same. A Lean countermodel shows that agreement of selected traces does not force an internal object morphism.

Status: exact logical obstruction to the inference; no internal morphism has been extracted.

3.3 Sigma(2,3,5) W1 finite structure

The checked finite surface includes:

- exact no-defect and component-7 alpha tables and admissibility tests;
- 30 end-node classes with explicit component fibres;
- exact trace aggregation and signs;
- an obstruction to a canonical residue action from representative-dependent S_3 transports;
- vanishing off-diagonal Hom spaces from disjoint momentum support;
- the intrinsic endomorphism algebra **C** $\times$ **C** on the two signed summands;
- exactly four intrinsic sign-pair S_3 actions, all nonfaithful;
- a separate four-dimensional multiplicity factor that creates nine S_3 types, four faithful, while remaining an added tensor factor rather than a source-intrinsic symmetry.

The generated family carrier and vertex-mode actions are finite algebraic shadows. They are not a construction of the full VOA.

3.4 Analytic and topological checks

Within their stated bounded or source-backed scopes, the following are checked:

- positivity and finite energy shells;
- rank-two oscillator grades and combined lattice–oscillator L_0 grades;
- finite diagonal traces and a nonzero g_b phase;
- twisted-module closure and the analytic trace envelope;
- prefactor/component consistency;
- an executable W1/Habiro recomputation showing that the manuscript token `H5` is a source-label defect and that the executable component is `H7`;
- the weakly negative plumbing prefactor $C_\Gamma = -q^{-5/4}$;
- unnormalised regularised totals $F_{000} = -\eta H_1$ and $F_{001} = -\eta H_7$;
- a calibrated orientation-reverse representative with a mandatory finite correction.

The physical Wilson half-index identification remains open. In the visual atlas it is deliberately marked by a dashed ?= rather than converted into a claim.

4 Mixed support and the K=4 raise

The newest exact bridge begins with the regularised indefinite-theta factor in *3d Modularity Revisited*. Away from its two cone boundaries, write

$$\epsilon_j(n) = \operatorname{sgn} B(c_j, n) \in \{-1, +1\}.$$

The pair (ϵ_1, ϵ_2) is a Klein four sign register under coordinatewise multiplication. The cone-support factor is exactly

$$\frac{1}{2}|\epsilon_1 - \epsilon_2| = \frac{1 - \epsilon_1\epsilon_2}{2}.$$

It is the XOR indicator: one on the two mixed-sign faces and zero on the two synchronised faces. This is an exact finite theorem, not a resemblance between two fourfold pictures.

4.1 Source-enriched selection of the second boundary

For the $6 - 1 = 5$ member of the source family, the displayed data are

$$A = \begin{pmatrix} -60 & 0 \\ 0 & 3 \end{pmatrix}, \qquad c_1 = (1, 0), \qquad c_2 = (3, 10).$$

Exact integer arithmetic gives

$$B(c_1, c_1) = -60, \quad B(c_2, c_2) = -240, \quad B(c_1, c_2) = -180.$$

The two primitive vectors therefore lie in the same negative-cone component. Their primitive perpendicular directions $(0, 1)$ and $(1, 6)$ have positive norms 3 and 48. Packet 198 proves this support skeleton but, using only A and c_1, correctly finds five source-form candidate rays.

Packet 200 retains the full ordered source datum

$$A = \operatorname{diag}(-2p\bar{p}, x), \qquad d_2 = (1, \bar{p}), \qquad \bar{p}x > 2p.$$

If $g = \gcd(x, 2p)$, the unique positive primitive integer generator of the A-orthogonal line $d_2^\perp$ is

$$c_2^{\text{prim}} = \left(\frac{x}{g}, \frac{2p}{g}\right).$$

The source vector $(x, 2p)$ is its positive g-fold multiple. Thus the full ordered cone datum canonically selects the projective c_2 ray. For the ordered Brieskorn data $(p_1, p_2, p_3) = (2, 3, 5)$, one has $p = 5$, $\bar{p} = 6$, $x = 3$, and hence $c_2 = (3, 10)$. The exact odd projector is

$$P_{c_2} = \begin{pmatrix} 9/4 & -3/8 \\ 15/2 & -5/4 \end{pmatrix},$$

with kernel $\mathbb{R}(1, 6)$ and image $\mathbb{R}(3, 10)$. It is idempotent, A-self-adjoint, and natural under exact bilinear coordinate changes.

There is also a source-scope correction and an exact follow-up. The paper states a broader $-\Sigma(2, 3, 6 \pm 1)$ cancellation, but the cited lemma directly assumes an $L = 4$ case. At $x = 3$ those hypotheses cover a $\Sigma(2, 3, 7)$-type ordering, not the displayed $L = 8\,\Sigma(2,3,5)$ datum. Packet 202 independently reconstructs the displayed no-defect $j = 2$ aggregate and isolates the unique coefficient

$$[q^{1/120}\bar{q}^{1/24}]\,\mathcal{S}^{(2)}_{235} = 2\exp(2\pi i/12) \neq 0.$$

The displayed no-defect aggregate therefore does not cancel. The pointwise c_2 support and source-enriched selector remain exact; all-sector and full $j = 1 + j = 2$ shadow questions remain open.

Status: exact XOR support theorem, exact source-enriched projective selector, exact arithmetic projector, and exact no-defect $j = 2$ noncancellation. No all-sector/full-shadow theorem, Jordan-algebra morphism, or physical half-index theorem is asserted.

4.2 Blind planes, ternary recovery, and the sphere

The project already contains exact finite models for the higher operation that motivated this comparison. In the two-weight model $W = \mathbb{C}_G \oplus \mathbb{C}_C$, the admissible equivariant cross-Hom spaces vanish: each complex plane is invisible to the other under the specified lower observer. In the shared-unit algebra J_κ, the mixed binary product vanishes while the ternary associator recovers the hidden factor. The obstruction is not discarded; it becomes operation data at the next rung.

On the centred zeta sphere, the critical great circle is $X = 0$ and

$$X_s = \frac{c_{-+}}{1 + |2s-1|^2}, \qquad c_{-+} = 2(2\,\mathrm{Re}\,s - 1).$$

Thus X is the compactified amplitude of the existing $(-,+)$ violation sector. A zero amplitude does not delete the support channel from the K=4 datum. The sphere is a geometric compactification of the complementary coordinate that a lower on-circle register cannot express internally.

Two different Klein four-groups act on this sphere: the holomorphic coordinate half-turn group and the classical zeta-zero reflection group. The current audit separates them. The zero quartet belongs to the latter; no conflation is used in the bridge above.

Inside this project, the "impossible extension" is the recursive predatum move to a higher obstruction language: a lower null relation or invisible support becomes a coordinate of the next stratum. The exact sign-register theorem supplies an external mixed-modular target, and the ordered source datum now selects its projective boundary ray. What remains open is a typed, algebra-valued natural map from the predatum/associator object to this arithmetic projector, together with all-sector/full-shadow and physical half-index checks.

The public formal archive also contains a rights-safe K-ladder charter integration. It preserves the intended shifted-rung vocabulary, zero-operator reading, and predatum test shape as project architecture, not as proved theorems. It explicitly separates the componentwise-XOR four-state register from the C_2 multiplicative unit subgroup and queues the missing typed intertwiner, the Hopf/no-global-section candidate, and a rung-by-rung dimension audit.

4.3 Cowlicks, sedenion annihilators, and coordinate half-turns

Packet 201 separates a useful topological ladder into its actual theorem levels. The classical Poincare-Hopf theorem forces a zero for every continuous tangent field on an even sphere. Bott-Milnor and Kervaire classify the positive-dimensional parallelizable spheres as S^1, S^3, S^7, and Adams gives the optimal number of independent tangent fields on every sphere. Those are cited external theorems.

The new project calculation is finite and fully replayable. In the audited Cayley-Dickson convention, every imaginary basis left-action matrix is skew and orthogonal in dimensions $2, 4, 8, 16$. The complete multiplication-frame identity holds in dimensions $2, 4, 8$ and fails in dimension 16. For the selected sedenion zero divisor

$$u = e_3 + e_{10},$$

the exact matrix L_u has rank 12 and kernel

$$\langle -e_5 + e_{12},\ e_4 + e_{13},\ e_7 + e_{14},\ -e_6 + e_{15} \rangle.$$

Thus the selected tangent field $x \mapsto ux$ on S^{15} vanishes on an S^3. This is not a claim that every field on the odd sphere must vanish or that this one example exhausts non-parallelizability.

On the centred compactified s-plane, the identity and three coordinate half-turns form K_4. The selected x-axis rotation field has zeros $\Sigma(0)$ and $\Sigma(1)$, each of local index $+1$; among the three coordinate-axis fields, only that pair lies off the critical great circle $X = 0$. Hairy-ball does not force exactly two zeros, the Radon-Hurwitz value is $\rho(16) = 9$ rather than 8, and no typed map identifies Adams's maximum of eight fields with a separate eight-dimensional ghost kernel.

Status: exact universal matrix identities, exact selected annihilator, exact coordinate K_4 and local-index calculation; classical topology cited; cross-program ghost/predatum/modular identifications open.

5 Descartes and split-zero geometry

Circle and Descartes geometry is secondary to the arithmetic and moonshine work in this release, but it supplies a useful exact geometric laboratory.

Packet 122 establishes a field-closure audit for the selected Descartes construction. Packet 123 supplies a reproducible D_3 orbit dataset with exact circle data and independent render coordinates. The visual archive contains the editable dataset, checker, GeoGebra companion, and depth-reveal assets. The depth-reveal family was rerendered in readability revision R2: larger high-contrast panel text, no clipped labels at 1920 x 1080, and an explicit 960 x 540 scaled-view gate. Its rebuilt storyboard, MP4, Blender scene, manifest, and handoff ZIP pass 22/22 checks.

Status: exact finite dataset and generated visualisation. No statement about an unbounded orbit or all source images follows from the selected depth.

The split-zero project title should be read as a research programme rather than a claim that every construction belongs to one completed axiomatic framework. Its most useful current role is to organise common deformation registers, complementary nullity conditions, and explicit failed or surviving maps.

6 Formal trust surface

The public Lean facade is intentionally small. It names the mathematical dependency chain without forcing readers to parse large enumeration tables. The complete formal archive contains the implementation modules, independent Python checkers, JSONL ledgers, and build logs.

The 17 July 2026 combined audit checks 31 theorem surfaces: 25 underlying statements and six readable facade aliases. Twenty-two report no axioms. Nine use only the standard Lean axioms `propext` and/or `Quot.sound`. No audited theorem depends on `native_decide` or a native code-generation axiom. The full Lake build completes successfully.

This is not advertised as globally axiom-free. The precise axiom footprint is included so that a reader can decide which proof paths meet their own trust requirements.

The central public aliases have the following footprint:

Alias	Lean reports
`acceptedFiniteRegisterIsComplete`	`propext`
`gamma107BlockDecomposition`	none
`sector289ActsByEightSixCycles`	none
`unnormalisedGamma107Sector289NoGo`	`propext, Quot.sound`
`degenerateKernelBridgeDiagnostic`	`propext, Quot.sound`
`generatedExponentialC6Repair`	none

7 What is not asserted

The present record does not assert any of the following:

- that matching q-series determine an internal moonshine/manifold morphism;
- that the unnormalised Γ_{107} is a six-step permutation action;
- that the generated exponential action is source-selected or unique;
- that the kernel-collapse diagnostic is a satisfactory bridge;
- that a finite carrier or mode table is a full VOA;
- that the two Klein-four actions on the centred zeta sphere are the same;
- that the K4/XOR support theorem alone selects $c_2 = (3, 10)$;
- that the displayed no-defect $L = 8$, $j = 2$ $\Sigma(2, 3, 5)$ aggregate cancels (Packet 202 proves that it is nonzero);
- that Packet 202's no-defect coefficient classifies every Wilson sector or the full $j = 1 + j = 2$ shadow;
- that any shadow simplification makes the second cone boundary dispensable;
- that the predatum support bridge proves a Wilson/half-index equality;
- that hairy-ball forces exactly two zeros or that S^{15} has no nonvanishing tangent field;
- that Adams's maximum of eight fields is identified with the ghost-kernel dimension without a typed map;
- that a bounded Descartes dataset proves an all-depth theorem;
- that conversational similarities are conjectures;
- that JT-authored source material is owned by, or relicensed through, this record.

8 Open research programme

The following questions survive the current checks and are stated because there is positive mathematical content behind them.

8.1 Source-natural exponential bridge

Can an external source action and a natural normalisation select a nondegenerate bridge compatible with the exact 48-state register? The generated E model proves finite feasibility, while the unnormalised no-go theorem and kernel diagnostic sharply constrain an acceptable answer.

8.2 Internal object relation behind common series

What additional structure, beyond equality of selected output traces, could support a genuine relation between the E_8^3 moonshine object and the $\Sigma(2, 3, 5)$ sector? The current result is an output-level equality plus an obstruction to inferring an internal map for free.

8.3 Predatum and complementary nullity

The project now has exact local models for the recursive operation: vanishing cross-Hom spaces between two weighted complex planes, a Hopf quotient with no global section, binary-null but ternary-recoverable mixed directions, a K=4 support register, and the XOR cone selector above. Together they establish a coherent finite mechanism in which a lower impossibility becomes a coordinate or operation datum of the next stratum.

The source-enriched arithmetic datum now selects $[c_2]$ and supplies a coordinate-natural odd projector. The global architecture remains open: Packet 199 finds standard Jordan, normal/conormal, affine, scheme-theoretic, operator-algebraic, and $\mathbf{F}_1$-adjacent constructions, but no natural algebra-valued morphism from the predatum/associator object to the arithmetic projector. The right status is therefore theorem-level local mechanism, source-enriched projective selector, and open cross-theory intertwiner.

9 Related Zenodo records

Earlier records remain useful as citable, versioned studies. They are not duplicated wholesale in this release.

Topic	Cite-all-versions DOI	Current role
Split-zero/common deformation project	10.5281/zenodo.20822444	Canonical public home; this record.
Full divisor locks	10.5281/zenodo.20207306	Prior Erdos–Strauss divisor-lock work.
Odd n-line work	10.5281/zenodo.20208683	Prior ES line decomposition.
Residual cubic support	10.5281/zenodo.20401937	Prior residual-support calculation.
Mixed mock Virasoro	10.5281/zenodo.20451738	Prior moonshine/modular study.
Rank-fifteen/Ogg study	10.5281/zenodo.20217934	Related finite/moonshine study.
Sedenion ghost study	10.5281/zenodo.20033645	Historical precursor to the checked target preflight.
Constructive terminal algebra	10.5281/zenodo.20376531	Historical exploration; not the current project home and not relied on as a present theorem surface.

10 Credit and citation

The project combines conceptual direction, source discovery, generated formalisation, exact computation, and visual explanation. The public record credits Jacolm Tobley as a researcher/project relationship without transferring ownership of JT-authored material. Generated model prose is never stored as JT verbatim.

For the evolving project, cite the concept DOI:

> The Clankers, *Split-Zero Geometry and Common Deformation Registers: Project Atlas, Exact Results, Formalization, and Visualizations*, Zenodo, 2026. https://doi.org/10.5281/zenodo.20822444.

For a fixed computational state, cite the version DOI shown on the Zenodo landing page together with the relevant stable theorem, packet, or certificate ID.